Bank Credit Limits, Alternative Finance, and Firm Investment

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Abstract

How does the cost of accessing alternative finance affect investment and output for a firm facing a bank-credit limit? A firm finances capital with internal funds, relatively cheap bank credit up to a limit, and an alternative source carrying an access wedge. The model derives a threshold equal to the discounted marginal product at the bank limit minus the bank rate. Above this threshold, alternative finance is too expensive and the bank limit determines investment. Below it, the firm invests beyond the limit until marginal product equals the alternative-finance cost. A larger bank line then changes the financing mix but not total investment, while a lower access wedge raises investment and output. A sectoral extension lets firms draw access wedges from sector-specific distributions, so the constrained share is generated by the model rather than assigned directly. Preliminary Peruvian forecast evidence motivates the exercise, but the model does not derive forecast accuracy, causality, or exact horizons.

Keywords: credit limits; alternative finance; investment; financing substitution; sectoral heterogeneity.
JEL codes: E22, E32, E44, G21, G32.

1 Introduction

Consider a firm that has exhausted its bank-credit line but still has a productive investment opportunity. It can stop at the bank limit or pay more to raise funds elsewhere. The relevant question is not simply whether another source exists. It is whether the return to the next unit of capital is high enough to cover the additional cost of accessing that source.

This paper asks: how does the cost of accessing alternative finance affect investment and output for a firm facing a bank-credit limit? The question distinguishes two objects often treated as interchangeable. Bank-credit availability determines how far a firm can invest at the cheap rate. The access wedge determines whether it is profitable to continue beyond that limit.

The model contains one firm and one continuous access wedge. The firm has internal funds, bank credit at rate R_B up to $\bar{b}$, and alternative finance at cost $R_B + \tau$. Production is increasing and strictly concave. The firm's marginal financing cost is therefore step-shaped, while its marginal product declines with installed capital.

The main result is a threshold proposition. Let $\bar{k} = n + \bar{b}$ be capital at the bank limit. For a firm that would like to invest beyond $\bar{k}$ at the bank rate, the critical access wedge is

$$\tau^* = \beta F'(\bar{k}) - R_B.$$

When $\tau \geq \tau^*$, alternative finance is too expensive and the firm stops at $\bar{k}$. When $\tau < \tau^*$, the firm invests beyond the bank limit until discounted marginal product equals $R_B + \tau$. The policy is continuous but kinked at the threshold.

Heterogeneity and sectors enter only after the firm result is established. Firms draw access wedges from a sector-specific distribution G_s. The share whose investment is governed by the bank limit is then $1 - G_s(\tau^*)$. This replaces the earlier practice of assigning a constrained share directly and makes the empirical assumption explicit: sectors differ in the distribution of access costs, not because every firm in a sector is one financing type.

Fernandez and Chavez (2026) provide the motivation for studying this heterogeneity. Their preliminary Peruvian forecasting exercise suggests that credit volume and a credit spread may contain different information across tradable and non-tradable activity. That pattern is not the object proved in this paper. Observed credit volume is an imperfect proxy for credit availability, and the measured bank spread is an imperfect proxy for the marginal financing cost. A static firm problem also cannot rank forecast horizons or establish causality.

The contribution is therefore deliberately narrow: a transparent access-cost threshold, its investment and output comparative statics, and a distributional bridge to sectoral heterogeneity. This question matches what the model can prove by hand, reproduce numerically, and formalize in Lean.

2 Related literature and positioning

The first strand studies equilibrium credit rationing. Stiglitz and Weiss (1981) show that asymmetric information can prevent the interest rate from clearing the loan market, so observationally similar borrowers may face a quantity restriction. Petersen and Rajan (1994) find that lending relationships benefit small firms primarily through greater availability of funds rather than large price reductions. The quantity regime below takes a credit ceiling as a reduced-form representation of those frictions; it does not reproduce their screening or relationship formation models.

The second strand studies balance-sheet and collateral amplification. Kiyotaki and Moore (1997) make durable assets both productive inputs and collateral, generating persistent responses through endogenous credit limits. Bernanke, Gertler, and Gilchrist (1999) embed an external-finance premium and heterogeneous capital-market access in a dynamic macroeconomic framework. Holmstrom and Tirole (1997) link firm net worth and intermediary capital to the supply of loanable funds. The present paper treats net worth and the credit ceiling as predetermined in its baseline. Section 8 shows how an endogenous ceiling can recover an accelerator without obscuring the main regime result.

The third strand establishes firm heterogeneity. Gertler and Gilchrist (1994) document that small firms contract more sharply after monetary tightening, while Petersen and Rajan (1994) emphasize their dependence on institutional lenders. These findings motivate a higher constrained share among non-tradable firms. This is a composition assumption to be disciplined by data, not a theorem that every small firm is non-tradable or every tradable firm has market access.

The fourth strand concerns leading financial indicators. Gilchrist and Zakrajsek (2012) construct a credit-spread measure with predictive power for future activity, and Faust et al. (2013) find that credit spreads improve real-time activity forecasts. Stock and Watson (2003) caution that predictive performance can be unstable across countries and periods. For Peru, Lahura (2011) finds a stable long-run credit-output relationship, while Butron and Pozo (2025) document a positive relationship using sectoral and regional heterogeneity.

The closest paper for the sectoral motivation is Muller and Verner (2024). They show that credit allocation toward non-tradable rather than tradable sectors predicts different macroeconomic outcomes, discuss smaller and more constrained non-tradable firms, and report robustness involving bond issuance. That evidence means neither the tradable/non-tradable split nor financing hetero-

geneity can be claimed as novel here. Holmstrom and Tirole's (1994) working-paper version and their published 1997 model also already contain capital thresholds associated with monitored and direct finance. Gilchrist, Sim, and Zakrajsek (2014) subsequently model the joint dynamics of investment and credit spreads.

The narrower course-paper contribution is to derive an explicit access-cost threshold within a capped-bank financing problem and use it to generate the constrained share from a distribution of firm wedges. The threshold separates a real quantity response from pure financing substitution. This is a provisional positioning claim, not a claim that credit rationing, external finance, or the broader empirical sector pattern is new.

3 Environment

There are two dates, t and $t+1$. A firm enters date t with internal funds $n > 0$. It chooses capital k, bank borrowing b, and alternative finance m. Capital becomes productive at $t+1$, linking the current financing decision to next-period production.

3.1 Technology

The general model uses a differentiable production function $F : \mathbb{R}_+ \to \mathbb{R}_+$.

Assumption 1 (Technology)**.** *$F(0) = 0$, $F'(k) > 0$, and $F''(k) < 0$ on the economically relevant domain. Future revenue is $F(k)$ and its date-t value is $\beta F(k)$, where $\beta \in (0, 1]$.*

Closed forms and the Lean verification use

$$F(k) = ak - \frac{c}{2}k^2, \qquad a > 0, \quad c > 0, \tag{1}$$

restricted to $k < a/c$, where output is increasing. This quadratic form is a local second-order representation of any smooth concave technology and keeps all regime boundaries transparent.

3.2 Financing

The firm's sources satisfy

$$k = n + b + m, \qquad 0 \le b \le \bar{b}, \qquad m \ge 0. \tag{2}$$

Bank finance has gross marginal cost $R_B > 0$. Alternative finance costs

$$R_M(\tau) = R_B + \tau, \qquad \tau \ge 0. \tag{3}$$

The access wedge τ includes issuance costs, information costs, and other barriers to non-bank funding. A very high wedge makes the alternative source economically irrelevant without requiring a separate binary access type.

Assumption 2 (Funding order)**.** $0 < R_B \le R_M < \beta F'(0)$*. The firm uses internal funds first, bank credit second, and alternative finance last. The strict upper bound guarantees a positive desired scale. The equations $\beta F'(k) = R_B$ and $\beta F'(k) = R_M$ are assumed to have solutions in the economically relevant domain; strict concavity makes those solutions unique.*

Empirically, the observed corporate loan spread is only a candidate proxy for τ or R_M. The model does not assume that a bank spread literally equals the cost of bonds or foreign credit. Section 8 discusses this identification boundary.

4 The firm problem

The firm maximizes discounted profits:

$$\max_{k,b,m} \quad \beta F(k) - R_B b - R_M m \quad \text{s.t.} \quad (2). \tag{4}$$

Let the capital stock attainable after exhausting the bank line be

$$\bar{k} \equiv n + \bar{b}. \tag{5}$$

Because bank finance is weakly cheaper, the reduced problem is continuous and piecewise:

$$\Pi(k) = \begin{cases} \beta F(k) - R_B(k-n), & n \le k \le \bar{k}, \\ \beta F(k) - R_B\bar{b} - R_M(k-\bar{k}), & k > \bar{k}. \end{cases} \tag{6}$$

The left and right first-order conditions are

$$\beta F'(k) = R_B \qquad \text{(bank-finance margin)}, \tag{7}$$

$$\beta F'(k) = R_M \qquad \text{(alternative-finance margin)}. \tag{8}$$

Let k_B and k_M solve (7) and (8). Strict concavity makes both targets unique. Since $R_M \ge R_B$, declining marginal product implies $k_M \le k_B$.

Under (1),

$$k_B = \frac{\beta a - R_B}{\beta c}, \qquad k_M = \frac{\beta a - R_M}{\beta c}, \qquad k_M \le k_B. \tag{9}$$

Lemma 1 (Optimal capital with alternative finance)**.** *Under Assumptions 1–2, the unique optimal capital stock is*

$$k^* = \begin{cases} n, & k_B \le n, \\ k_B, & n < k_B < \bar{k}, \\ \bar{k}, & k_M \le \bar{k} \le k_B, \\ k_M, & \bar{k} < k_M. \end{cases} \tag{10}$$

Proof. On $[n, \bar{k}]$, the derivative of profit is $\beta F'(k) - R_B$; on $(\bar{k}, \infty)$ it is $\beta F'(k) - R_M$. Both are strictly decreasing. If $k_B \le n$, the left derivative is already non-positive at n. If $n < k_B < \bar{k}$, it crosses zero at k_B before the kink. If $k_M \le \bar{k} \le k_B$, the derivative is weakly positive immediately to the left of the kink and weakly negative immediately to its right, so the kink is optimal. Finally, if $\bar{k} < k_M$, profit continues increasing after the bank line is exhausted and reaches its maximum at k_M. Strict concavity gives uniqueness, with adjacent formulas agreeing at equality boundaries. □

5 The access-cost threshold

Focus on the economically relevant case in which the firm would like to invest beyond the bank limit at the bank rate:

$$\beta F'(\bar{k}) > R_B. \tag{11}$$

Define the access-cost threshold

$$\tau^* \equiv \beta F'(\bar{k}) - R_B > 0. \tag{12}$$

Proposition 1 (Access threshold, investment, and output)**.** *Under Assumptions 1–2 and (11):*

(i) If $\tau \geq \tau^*$, alternative finance is too expensive and $k^* = \bar{k} = n + \bar{b}$.

(ii) If $0 \leq \tau < \tau^*$, the firm uses alternative finance and its unique optimum satisfies

$$\beta F'(k^*) = R_B + \tau, \qquad k^* > \bar{k}. \tag{13}$$

(iii) Within strict regimes,

$$\tau > \tau^* : \qquad \frac{\partial k^*}{\partial \bar{b}} = 1, \qquad \frac{\partial k^*}{\partial \tau} = 0, \tag{14}$$

$$0 \leq \tau < \tau^* : \qquad \frac{\partial k^*}{\partial \bar{b}} = 0, \qquad \frac{\partial k^*}{\partial \tau} = \frac{1}{\beta F''(k^*)} < 0. \tag{15}$$

Output $y_{t+1} = F(k^*)$ has the corresponding signs by the chain rule.

Proof. At $\bar{k}$, the derivative of profit immediately to the left is positive by (11). The right derivative is

$$\Pi'_+(\bar{k}) = \beta F'(\bar{k}) - (R_B + \tau) = \tau^* - \tau.$$

If $\tau \geq \tau^*$, the slope changes from positive to non-positive at the kink, so strict concavity makes $\bar{k}$ the unique optimum. If $\tau < \tau^*$, profit continues increasing beyond $\bar{k}$ and strict concavity gives the unique solution to (13). Implicitly differentiating that condition yields $\beta F''(k^*)\, \partial k^*/\partial \tau = 1$. The remaining derivatives follow directly from $k^* = n + \bar{b}$ in the constrained regime and from the independence of (13) from $\bar{b}$ in the alternative-finance regime. □

For the quadratic technology,

$$\tau^* = \beta\big[a - c(n + \bar{b})\big] - R_B, \qquad k_M(\tau) = \frac{\beta a - R_B - \tau}{\beta c}, \tag{16}$$

so, within the focal domain,

$$k^*(\tau) = \max\{n + \bar{b}, k_M(\tau)\}, \qquad \frac{\partial k_M}{\partial \tau} = -\frac{1}{\beta c}. \tag{17}$$

More directly, the amount invested beyond the bank limit is

$$k^*(\tau) - \bar{k} = \max\left\{0, \frac{\tau^* - \tau}{\beta c}\right\}. \tag{18}$$

Thus the distance below the threshold maps linearly into additional capital. If $\Delta k \equiv k^* - \bar{k}$, the associated output gain is

$$F(k^*) - F(\bar{k}) = \Delta k\big[a - c\bar{k}\big] - \frac{c}{2}(\Delta k)^2 > 0 \tag{19}$$

whenever $\Delta k > 0$ and both capital stocks remain in the increasing region.

Remark 1 (Local zero effects). The zero derivatives are regime statements, not universal claims. A sufficiently large change in τ can move a firm across τ^*, and a sufficiently large increase in the bank ceiling changes the threshold itself. The model therefore predicts a continuous but kinked policy rather than universal zero effects.

6 Sectoral aggregation and empirical mapping

Let firms differ in their access wedges. In sector $s \in \{N, T\}$, let τ_i have a continuous cumulative distribution G_s. Holding the remaining primitives fixed, Proposition 1 generates the constrained share:

$$\lambda_s \equiv \Pr(\tau_i \geq \tau^*) = 1 - G_s(\tau^*). \tag{20}$$

This is an outcome of the threshold model, not a separate assignment of firms to binary financing types.

Sectoral output can be written as

$$Y_{s,t+1} = \int_0^{\tau^*} F\big(k_M(\tau)\big)\, dG_s(\tau) + \big[1 - G_s(\tau^*)\big] F(\bar{k}). \tag{21}$$

The first term collects firms that use alternative finance; the second collects firms stopped by the bank limit. Because $k_M(\tau^*) = \bar{k}$, output is continuous at the boundary.

If the non-tradable distribution is shifted toward higher access wedges, then $1 - G_N(\tau^*) > 1 - G_T(\tau^*)$: more non-tradable firms have investment pinned to bank availability. A distribution shifted toward lower wedges contains more firms whose scale is governed by the cost of alternative finance. This is a testable hypothesis about G_N and G_T, not a theorem that every firm in one sector is constrained.

The mechanism suggests three empirical checks. First, credit-quantity effects should be stronger among firms with exhausted bank lines and high access wedges. Second, financing-cost effects should be stronger among firms already using bonds, foreign credit, or other non-bank sources. Third, a larger bank line should raise total finance for firms above the threshold but mainly change debt composition for firms below it. These interactions are sharper tests than a mechanical mapping from the static model to forecast horizons.

7 Numerical illustration

Consider $\beta = 0.95$, $a = 2$, $c = 0.5$, $n = 0.4$, $\bar{b} = 0.5$, and $R_B = 0.5$. Then

$$\bar{k} = 0.9, \qquad k_B = 2.947, \qquad \tau^* = 0.9725. \tag{22}$$

With $\tau = 1.10$, alternative finance is too expensive and the firm stops at $k = 0.9$. With $\tau = 0.90$, the alternative-finance target is $k = 1.053$, so the firm uses 0.153 of alternative finance after exhausting its bank line.

Table 1: Illustrative regime responses

Experiment	$\tau = 1.10$	$\tau = 0.90$
Baseline capital	0.900	1.053
Baseline output	1.598	1.828
$\bar{b}$ rises by 0.10	capital rises to 1.000	capital unchanged
τ rises by 0.02	capital unchanged	capital falls to 1.011
Relevant margin	bank-credit availability	alternative-finance cost

For an illustrative sectoral exercise, let non-tradable wedges be uniform on $[0.8, 1.2]$ and tradable wedges uniform on $[0.6, 1.0]$. Equation (20) then gives constrained shares of 0.569 and 0.069, respectively. These distributions are chosen to illustrate the mapping, not calibrated to Peru.

8 Extensions and limitations

Endogenous bank ceiling. Let $\bar{b}_{i,t} = \phi_s n_{i,t}$, with pledgeability $\phi_s > 0$. In the constrained regime, $k_{i,t} = (1 + \phi_s) n_{i,t}$, so a net-worth shock changes investment directly and through borrowing capacity. Adding $n_{i,t+1} = \omega \Pi_{i,t}$ creates a simple financial accelerator. This extension is useful for propagation, but not necessary for the core access-threshold proposition.

Finite alternative-finance capacity. The baseline lets a firm raise any amount at marginal cost $R_B + \tau_i$. A finite limit $\bar{m}_i$ would add a second quantity constraint. The current paper omits it because the empirical question concerns whether the first unit beyond the bank limit is financed; adding another cap would not change the threshold for that first unit.

Dynamic capital and horizons. A disciplined dynamic version introduces $K_{t+1} = (1 - \delta) K_t + I_t$, a convex installation cost $\Phi_s(I_t)$, and the same two financing tranches for investment. Let $q_{i,t}$ denote the current-value shadow price of installed capital. The investment and capital first-order conditions imply

$$q_{i,t} = R_{j,t} + \Phi_s'(I_{i,t}) + \mu_{i,t}, \tag{23}$$

$$q_{i,t} = \beta \mathbb{E}_t \left[F_s'(K_{i,t+1}) + (1 - \delta) q_{i,t+1} \right], \tag{24}$$

where $j = B$ when the marginal unit is bank financed and $j = M$ when it uses alternative finance. The term $\mu_{i,t} \geq 0$ is the shadow wedge attached to a binding quantity limit; it is zero away from that limit (with the bank/market cost gap instead determining which tranche is marginal). Combining (23)–(24) yields the Euler equation

$$R_{j,t} + \Phi_s'(I_{i,t}) + \mu_{i,t} = \beta \mathbb{E}_t \left[F_s'(K_{i,t+1}) + (1 - \delta) \{ R_{j',t+1} + \Phi_s'(I_{i,t+1}) + \mu_{i,t+1} \} \right]. \tag{25}$$

Thus a credit-quantity innovation operates through the current and expected shadow wedges, whereas an access-cost innovation changes the marginal financing price directly. Sector-specific curvature in Φ_s, depreciation, or project gestation lags can shift the output response across horizons and may rationalize why preliminary quantity evidence appears at relatively short horizons while price information appears over a longer horizon. Estimating those dynamic parameters is left outside the current proposition so that the first deliverable remains exactly provable by hand; equation (25) states the disciplined route for the next version rather than claiming to identify the observed horizons already.

Measurement. The model's τ is an access wedge, while the thesis uses a bank loan spread assigned by firm size because sector-level rates are unavailable. The empirical proxy may therefore contain measurement error and need not move one for one with bond or foreign-credit costs. A stronger empirical design would test whether the spread co-moves with funding substitution or use direct measures of bond, foreign-credit, or non-bank access.

Boundaries of the result. The model abstracts from default, risk choice, lender optimization, general equilibrium, and feedback from future expected output to current credit supply. It explains a conditional transmission mechanism, not the origin of the credit ceiling or the causal identification of credit shocks.

9 Conclusion

The cost of alternative finance determines whether a bank-credit limit becomes a real investment constraint. Above τ^*, the firm stops at the limit and additional bank availability raises investment and future output. Below τ^*, the firm finances the marginal unit elsewhere; a larger bank line changes the source of funds but not the scale of investment, while the access wedge governs that scale. Sectoral differences arise from distributions of τ_i, not from assigning every firm in a sector to one financing type.

This mechanism turns the thesis's preliminary sectoral pattern into a formal economic hypothesis with explicit assumptions, first-order conditions, regime boundaries, and falsifiable cross-sectional predictions. Its value for the course is not that it proves the forecasting result, but that it supplies the firm problem the empirical exercise previously treated as a black box.

A Additional derivations

For the quadratic technology, profit on a segment with marginal funding cost R is, up to constants,

$$\pi_R(k) = \beta\left(ak - \frac{c}{2}k^2\right) - Rk. \tag{26}$$

The FOC and second-order condition are

$$\pi_R'(k) = \beta(a - ck) - R = 0, \qquad \pi_R''(k) = -\beta c < 0. \tag{27}$$

Completing the square gives the global identity

$$\pi_R(k_R) - \pi_R(k) = \frac{\beta c}{2}(k - k_R)^2 \geq 0, \tag{28}$$

where $k_R = (\beta a - R)/(\beta c)$. At the bank ceiling, global optimality in the kink regime follows from

$$\Pi(\bar{k}) - \Pi(k) = (\bar{k} - k)\big[\beta F'(\bar{k}) - R_B\big] + \frac{\beta c}{2}(\bar{k} - k)^2 \geq 0, \qquad k \leq \bar{k}, \tag{29}$$

$$\Pi(\bar{k}) - \Pi(k) = (k - \bar{k})\big[R_M - \beta F'(\bar{k})\big] + \frac{\beta c}{2}(k - \bar{k})^2 \geq 0, \qquad k \geq \bar{k}. \tag{30}$$

These are the identities formalized by the Lean theorem `kink_regime_optimal`.

B Cobb-Douglas robustness

If $F(k) = Ak^\alpha$ with $A > 0$ and $0 < \alpha < 1$, the two unconstrained targets are

$$k_j = \left(\frac{\beta\alpha A}{R_j}\right)^{\frac{1}{1-\alpha}}, \qquad j \in \{B, M\}. \tag{31}$$

The policy function retains exactly the four cases in Lemma 1. In the alternative-finance regime,

$$\frac{\partial k_M}{\partial \tau} = -\frac{k_M}{(1-\alpha)R_M} < 0, \qquad \frac{\partial F(k_M)}{\partial \tau} = -\frac{\alpha}{1-\alpha}\frac{F(k_M)}{R_M} < 0. \tag{32}$$

Thus the regime ranking is not an artifact of the quadratic technology.

C Lean formalization status

Table 2 records the declaration-level status of every numbered result. The current files in `lean/prototype/` are a direct Lean 4 prototype: they verify the main quadratic identities and sensitivity inequalities without `sorry`, but they are not yet the complete folder generated by the course's AppliedModelingLib workflow. Before the final paper, the merged paper PDF will be pinned to a commit, the workflow will be run on that version, and the entire generated folder will replace the prototype.

Table 2: Paper result to Lean declaration map

Paper result	Prototype declaration(s)	Current status
Lemma 1	`kink_regime_optimal`; `market_regime_optimal`	Core kink and market branches proved; full four-case wrapper pending AppliedModelingLib run.
Proposition 1	`access_below_threshold_iff`; `market_target_at_access_threshold`; `targetCapital_sub_cap_eq_threshold_gap`; `market_target_decreases`	Threshold equivalence, continuity, the investment-gap identity, and the direction of the quadratic access-cost response are proved. The general concave statement remains pending the AppliedModelingLib run.

Lean verifies mathematics conditional on supplied assumptions. It does not verify that those assumptions hold in Peruvian data, that the literature review is exhaustive, or that the empirical estimates identify causal effects. Paper commit formalized: **pending**; AppliedModelingLib `check -fast`: **pending**.

D AI collaboration log

AI assistance was used to search and compare the literature, propose the two-tranche financing model, check symbolic and numerical claims, draft LaTeX, and prototype Lean statements. The author retained responsibility for the economic question, the interpretation of the thesis evidence, and each accepted assumption. Raw prompts and relevant answers are preserved in `prompts.md`.

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Table 3: Claims checked during AI-assisted development

AI-assisted claim	Independent check	Verdict
An access wedge below τ^* implies that desired capital exceeds the bank limit.	Hand derivation, symbolic identity, numerical check, and Lean prototype.	Accepted for the stated concavity and positivity conditions.
A larger bank line does not change total capital below τ^*.	FOC, implicit derivative, and numerical check.	Accepted locally while alternative finance remains strictly positive; bank credit still affects profits and debt composition.
The inequality $\lambda_N > \lambda_T$ can be treated as the sectoral mechanism.	Re-derived the constrained share from firm wedges.	Rejected as circular. The revision uses $\lambda_s = 1 - G_s(\tau^*)$ and treats differences in G_s as the empirical hypothesis.
The model explains exact empirical forecast horizons.	Compared the static timing with the unfinished thesis results.	Rejected. The draft now states no claim about horizon rankings.
The sector pattern itself is novel.	Checked Muller and Verner (2024), its appendices, and related working papers.	Rejected. Novelty is limited to the explicit access-cost threshold and its sectoral distributional mapping.

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